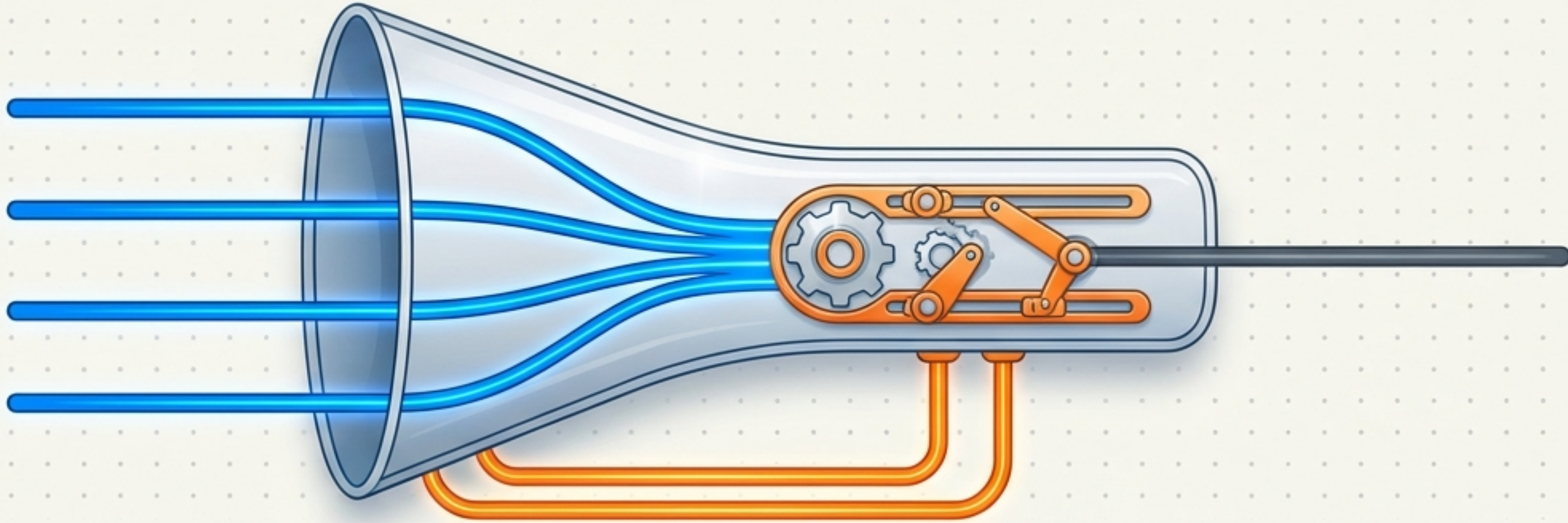
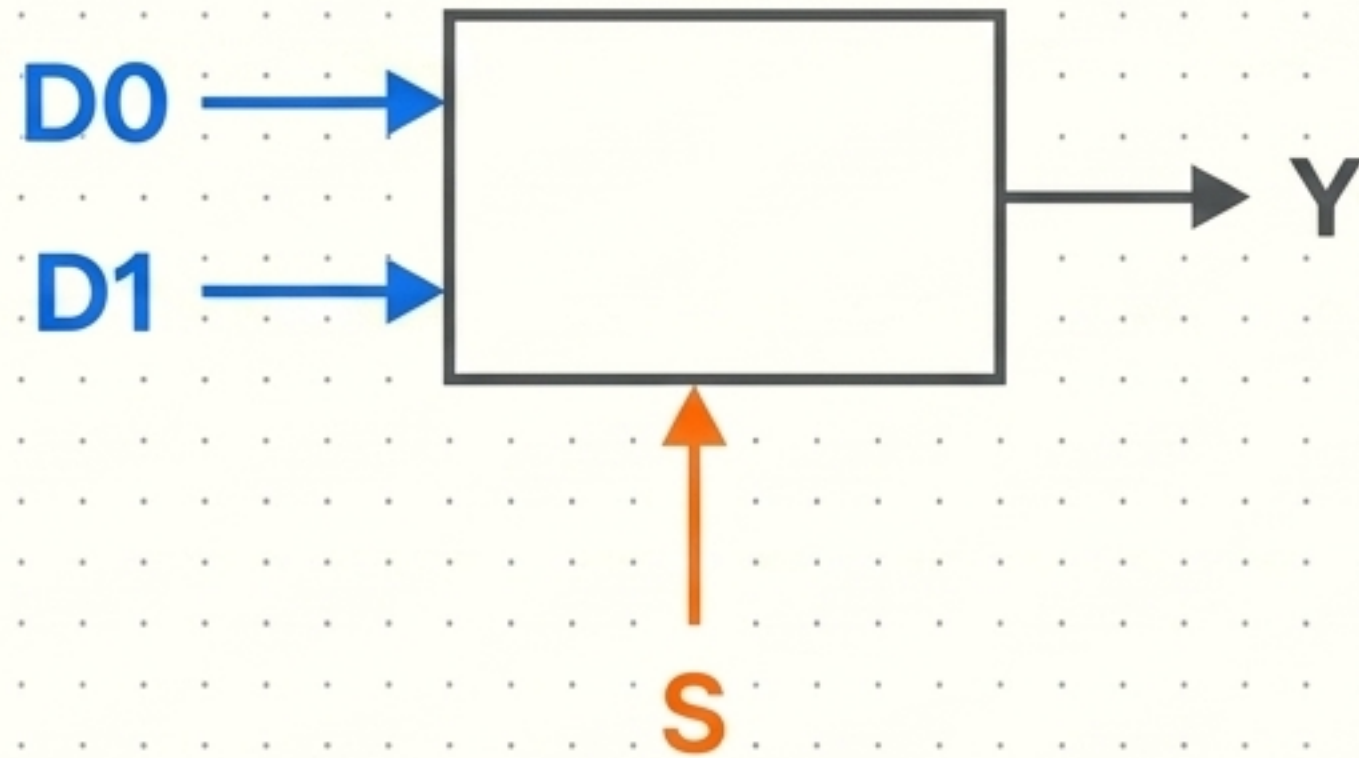


MULTIPLEXER MUX OVERVIEW

DIGITAL DATA SELECTOR



- **2 POWER N INPUT LINES DATA**
 - **N SELECT LINES CONTROL**
 - **EXACTLY 1 OUTPUT LINE**



2 TO 1 MUX

$$Y = S' D0 + S D1$$

$S = 0$ LEADS TO $Y = D0$

$S = 1$ LEADS TO $Y = D1$

NUMERICAL EXECUTION

FIND Y WHEN $D0 = 0$, $D1 = 1$, $S = 1$

STEP 1

EQUATION: $Y = S' D0 + S D1$

STEP 2

SUBSTITUTION: $Y = (1)' (0) + (1) (1)$

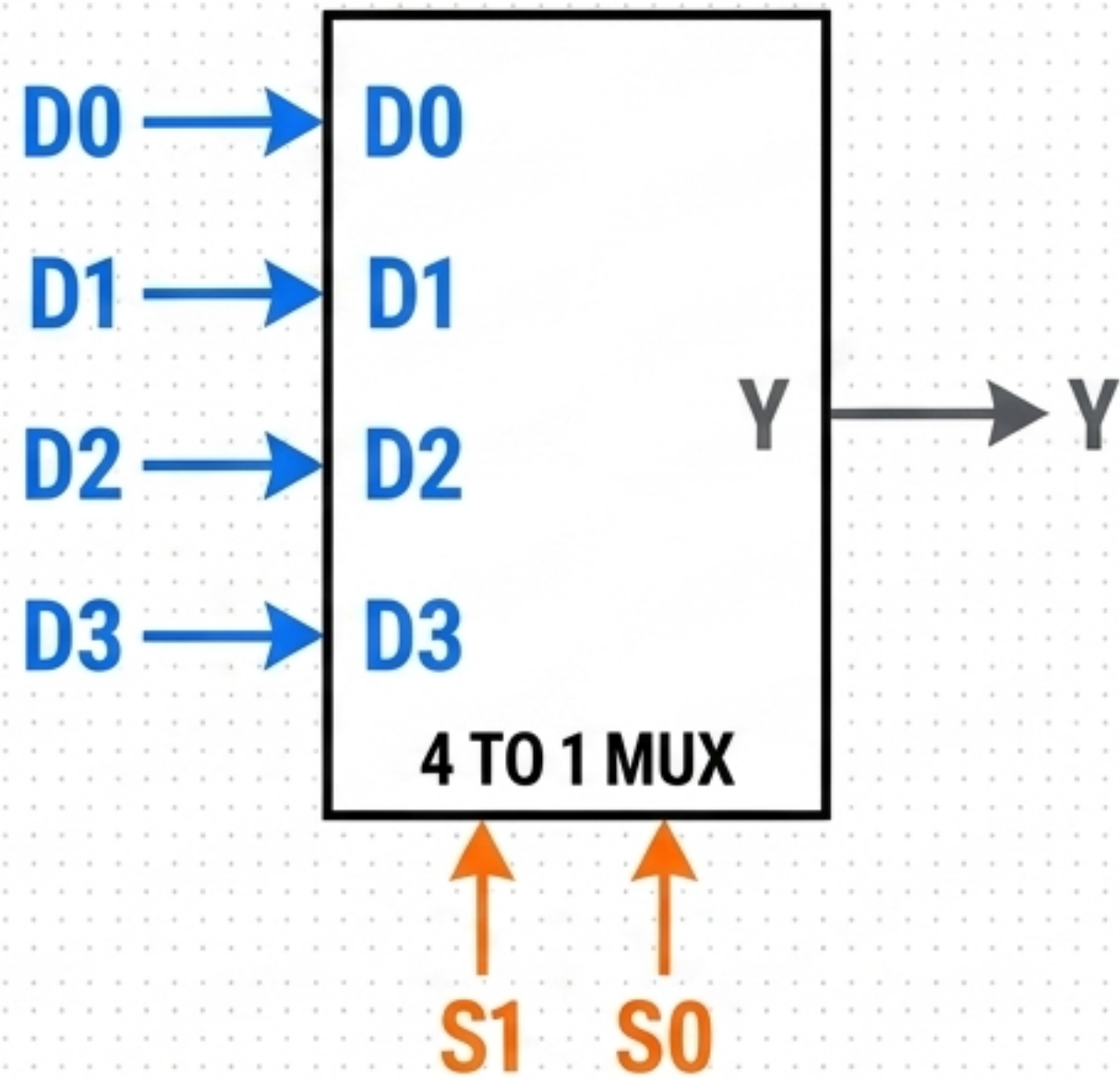
STEP 3

ARITHMETIC: $Y = (0 * 0) + (1 * 1)$

STEP 4

SOLVE: $Y = 0 + 1 = 1$

FINAL RESULT: $Y = 1$



4 TO 1 MUX

$$Y = S1' S0' D0 + S1' S0 D1 + S1 S0' D2 + S1 S0 D3$$

NUMERICAL EXECUTION

FIND Y WHEN $D0 = 1$, $D1 = 0$, $D2 = 1$, $D3 = 0$, $S1 = 1$, $S0 = 0$

STEP 1 EQUATION

$$Y = S1' S0' D0 + S1' S0 D1 + S1 S0' D2 + S1 S0 D3$$

STEP 2 SUBSTITUTION

$$Y = (1)' (0)' (1) + (1)' (0) (0) + (1) (0)' (1) + (1) (0) (0)$$

STEP 3 ARITHMETIC

$$Y = (0 * 1 * 1) + (0 * 0 * 0) + (1 * 1 * 1) + (1 * 0 * 0)$$

STEP 4 SOLVE

$$Y = 0 + 0 + 1 + 0 = 1$$

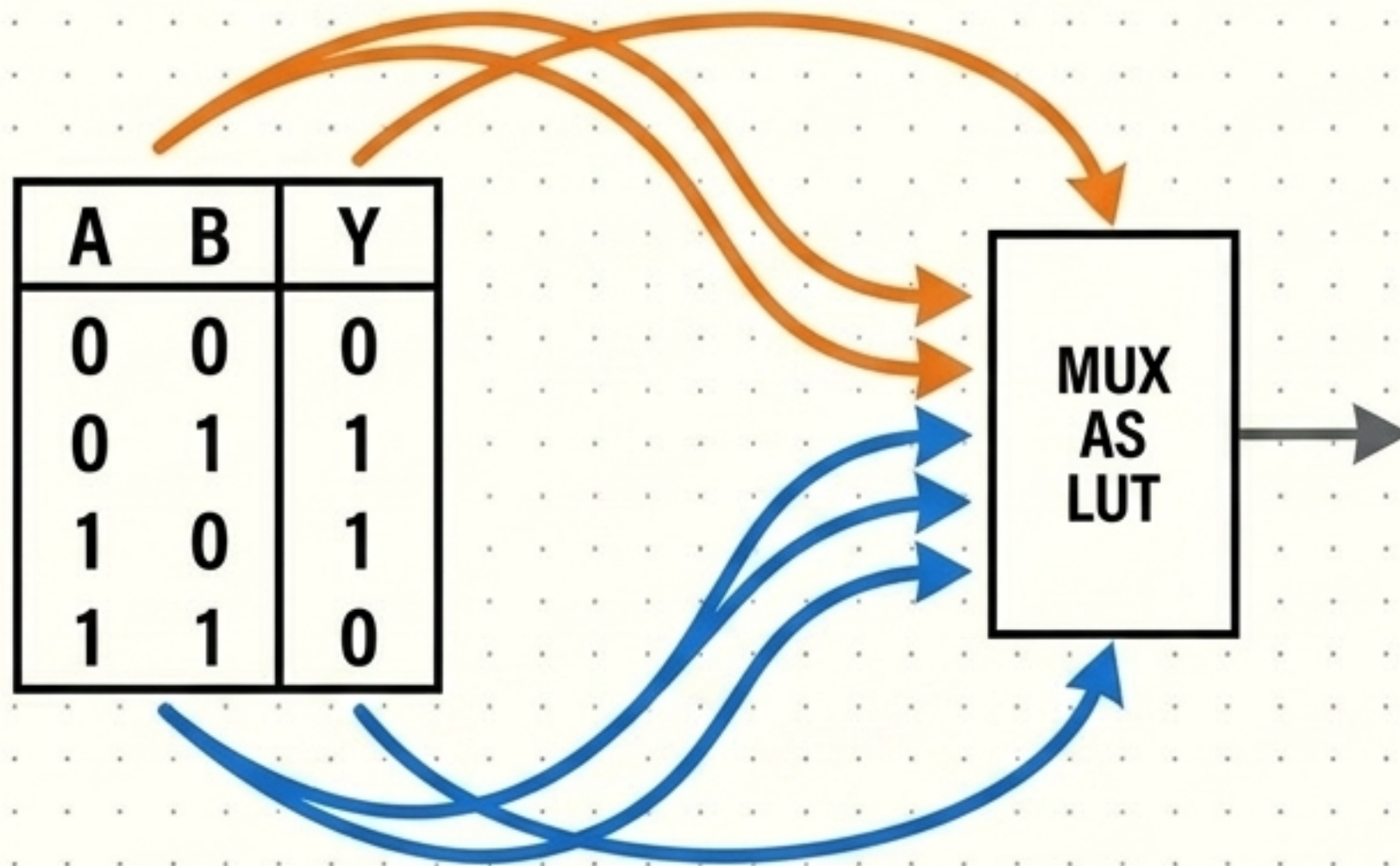
FINAL RESULT: Y = 1

LOGIC SYNTHESIS

MUX AS LUT

ANY N VARIABLE LOGIC GATE CAN BE BUILT USING A MUX

LOGIC INPUTS MAP TO MUX SELECT LINES



TRUTH TABLE OUTPUTS MAP TO MUX DATA INPUTS 0 OR 1

NUMERICAL EXECUTION XOR GATE

BUILD 2 INPUT XOR GATE A XOR B USING 4 TO 1 MUX

STEP 1

SET SELECT LINES: $S_1 = A$, $S_0 = B$

STEP 2 MAP TRUTH TABLE TO DATA INPUTS:

$A = 0, B = 0$ GIVES OUT = 0 SO $D_0 = 0$

$A = 0, B = 1$ GIVES OUT = 1 SO $D_1 = 1$

$A = 1, B = 0$ GIVES OUT = 1 SO $D_2 = 1$

$A = 1, B = 1$ GIVES OUT = 0 SO $D_3 = 0$

STEP 3 WRITE EQUATION:

$$Y = A' B' (0) + A' B (1) + A B' (1) + A B (0)$$

STEP 4 SIMPLIFY:

$$Y = A' B + A B'$$

**FINAL RESULT: XOR GATE
SYNTHESIZED SUCCESSFULLY**